

A Replication of Study 3 in Stellar et al. (2018): Awe and Humility

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As part of the CREP project... we follow a direct replication and extension approach.

The seminal findings of the Open Science Collaboration (2015) triggered what is now known as the ‘replication crisis’ in psychological science, prompting researchers to question thusfar widely unchallenged publications by and large. Since then, there has been an ongoing effort within the psychological research community to actively pursue replication of affected studies with important findings, aiming to reevaluate their validity and ultimately restore trust in the body of psychological knowledge. As part of the Collaborative Replications and Education Project (CREP) project of Grahe et al. (2026), we aim to contribute to this endeavour by investigating the replicability of the results in ‘Study 3’ of Stellar et al. (2018): Awe and Humility. In this study, the authors hypothesize momentary experiences of awe to cause a temporary increase in humility. The authors operationalize this hypothesis by means of a randomized video intervention, designed to induce either a state of awe or a neutral state, with a subsequent evaluation on a behavioral measure of humility. Both theoretical background and original study design are outlined in detail in the original publication of Stellar et al. (2018). To ensure transparency, following the general recommendation of adopting the principles of open science (Schönbrodt et al. 2018), we have made all relevant materials for this replication study available on the platform Open Science Framework (OSF), in particular the preregistration form at <https://osf.io/537aq>.

A Direct Replication and Extension of Stellar et al. (2018)

In this replication study, we aim to perform a direct replication of study 3 Stellar et al. (2018), adopting the authors original methodology and analysis to the best of our knowledge and possibilities, while further – in addition – extending the study design to allow for analyses of supplementary hypotheses, taking care to ensure the original study remains intact as an isolated part of our extended study design. As our recruitment of participants has taken place in Germany, however, we have opted to translate the study material into German, in particular the audio track for the neutral condition video. Our recruitment process follows the

snowball sampling principle (Goodman 1961), so consequently our sample consists of mostly students, as well as their friends and relatives, resulting in demographic range that differs from to the original sample of Stellar et al. (2018). All known deviations are documented in detail within the preregistration form. For the direct replication, we aim to replicate the following hypothesis.

H1: Relative to a neutral control condition, the original awe inducing video increases humility in a behavioral measure.

We extend the replication study by introducing a third condition, in which participants are shown a more modern video, which is also designed to induce awe. Based on this condition, we additionally investigate the following confirmatory hypothesis.

H2: Relative to a neutral control condition, the new awe inducing video increases humility in a behavioral measure.

Furthermore, we aim to address the natural follow-up question of how the two experimental conditions compare by means of the following exploratory hypothesis.

E1: The new (more modern) awe inducing video leads to higher awe ratings compared to the old awe inducing video.

Note, that as documented in the preregistration, we have formulated a total of six supplementary exploratory hypotheses. Of these, E1 was chosen at random prior to any further data analyses. The full list of all exploratory hypotheses is documented in the publicly available preregistration form.

Method

The scripts required for reproducing the following results have been made publicly available on GitHub at https://github.com/TiziSchuh/Stellar_Rep_01. The hypotheses, operationalizations, analysis plan, inference criteria, and sample size justification have been preregistered at <https://osf.io/537aq>. Open material and data can all be found at <https://osf.io/cy287>.

Participants

The initial sample had 189 participants. We preregistered to exclude participants who did not write anything in the writing assignment (26 exclusions) or indicated that their data should not be used (1 exclusions), leading to a final sample size of $n = 162$, resulting in conditions being distributed 50 (neutral video), 58 (original awe video) and 54 (modern awe video). The final sample had a mean age of 32 ($SD = 16.2$, range: 18–77). 112 participants reported identifying as female, 46 as male and 3 as non-binary, with 1 participant opting not to report their gender identity.

Measures and Materials

We operationalized our hypotheses by measuring three variables. For the Independent Variable (IV), we measured the randomized video condition with levels “neutral”, “awe original” and “awe modern”. We then measured awe as an emotional state on a behavioral scale, computed as the mean of three items (“Staunen”, “Faszination” and “Ehrfurcht”), encoding responses on a seven point likert scale each.

For the Dependent Variable (DV), we measured the state of humility as the log-transformed ratio of listed personal strengths to weaknesses. The number of strengths and weaknesses are counted by two independent raters from the free-text answer of the participants in a writing exercise. We use the “Codebook Instructions for Counting Participant-Reported Strengths and Weaknesses” from <https://doi.org/10.26443/msurj.v19i1.231> (see Appendix B).

Link to code-book!!!

Procedure

We have provided an overview in Figure 1.

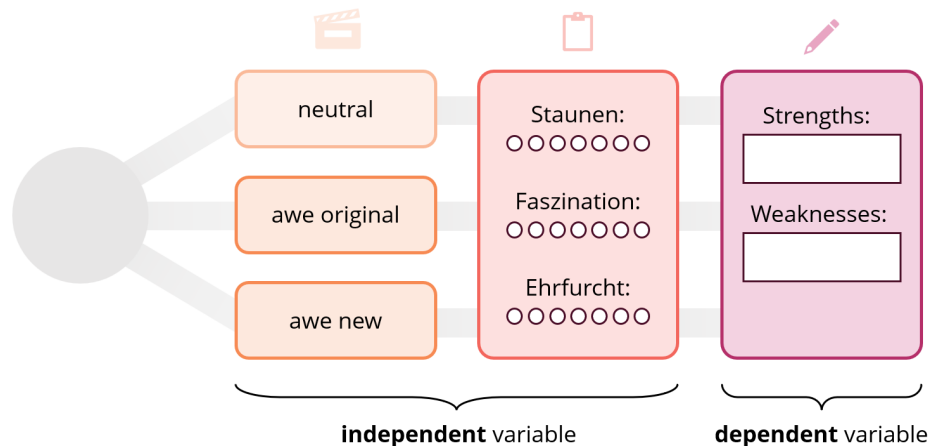


Figure 1: *An Illustration of the Study Procedure.*

One-time, cross-sectional online study. Data collection: 19.06.2026 - 10.07.2026. Participants first watch the video, then do the strength/weakness writing exercise, then report on their emotional state. Finally they report on the control questions.

Statistical Procedure

Exclude persons with zero weakness to calculate log. Through log we get more normally distributed data. For H1 and H2, we perform t-test with a p-value of...

Deviations from Preregistration

Power not enough bc valid participants?

Results

Confirmatory Results

The results was: $\Delta M = 0.09$, 95% CI $[-0.10, \infty]$, $t(71) = 0.80$, $p = .212$.

Exploratory Results

The new video... (just as effective on awe measure, effect also non-significant)

Discussion

Limitations and Future Directions

lack of evidence is not evidence of absence. Maybe just power not big enough or unlucky... but for this: CREP. Also, we have not tested for MI, explicitly investigating to what extent the translation may have an impact on the reproducibility of the effect would adress this limitation.

Conclusion

References

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Schönbrodt, Felix D, Markus Maier, Moritz Heene, and Markus Bühner. 2018. “Forschungstransparenz Als Hohes Wissenschaftliches Gut Stärken.” *Psychologische Rundschau*.

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This Section Is an Appendix

Another Appendix